GlycoViz User Manual

Installation:

1. Install docker-compose on your machine. https://docs.docker.com/compose/install/
2. Download GlycoViz code from https://github.com/calico/glycoviz
3. In the root directory, run the command "docker-compose up".
4. The web app should accessible from http://localhost:5173

Manual:

The workflow consists of three stages: (1) configure and upload, (2) run analysis, and (3) explore results.

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1. Data Upload and Configuration

1.1 Uploading a Data File

From the home page, drag and drop a search engine results file into the upload zone, or click to browse. Supported formats include .xlsx, .tsv, and .csv. GlycoViz automatically detects the search engine format (Byonic, MSFragger, or your customized headers) by matching column headers against the configured header sets. The detected format is displayed below the upload zone. All input file headers can be self-defined in the UI.

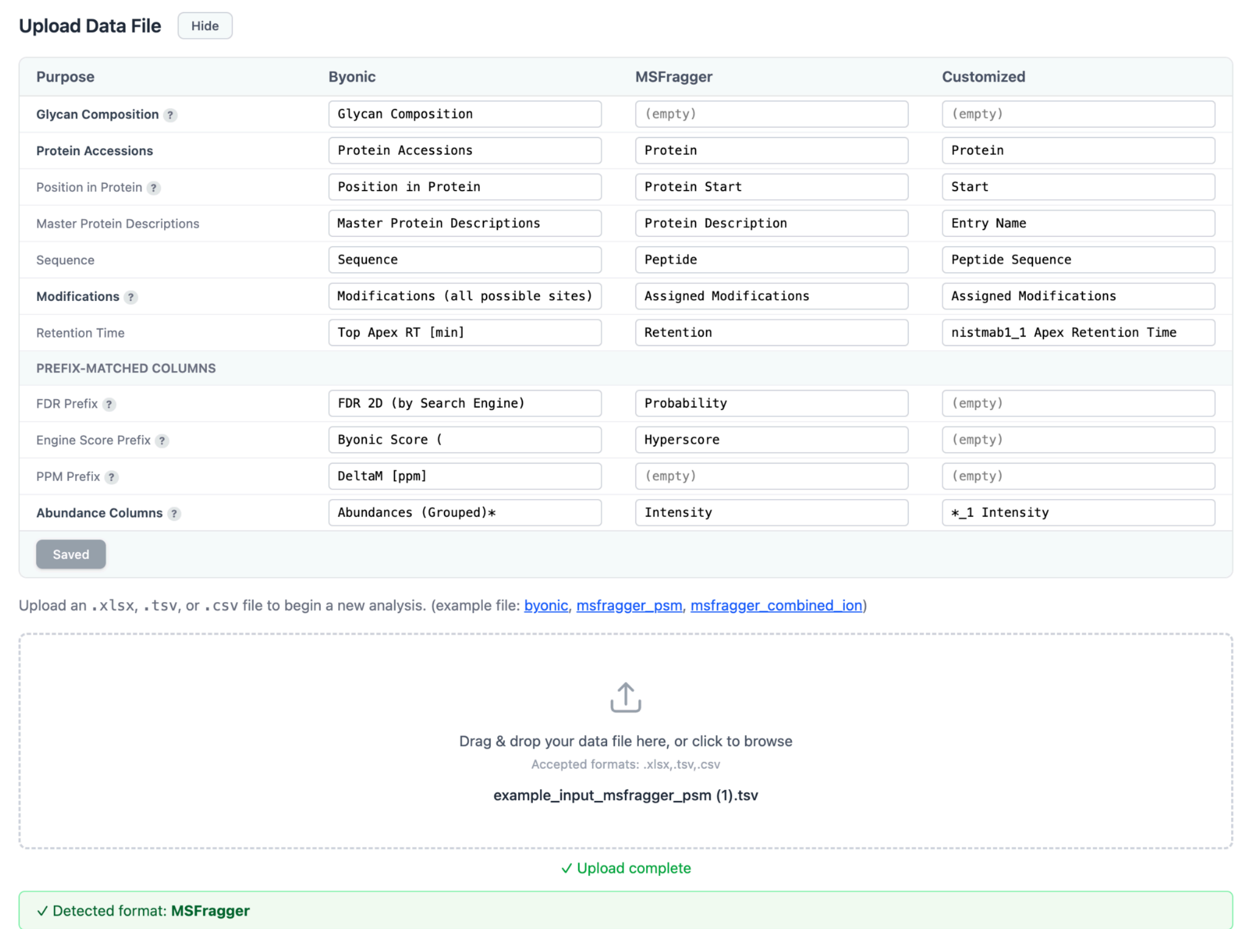
1.2 Column Header Settings

Column Settings table defines how GlycoViz maps your file's column headers to internal field names. Three presets are provided:

- Byonic — default mapping for Byonic search results
- MSFragger — mapping for MSFragger/FragPipe output
- Customized — a user-editable template for other search engines

Each row corresponds to a required field. Required fields are bold in the table. See tooltips in the UI for instructions. The Abundance Columns field supports * wildcards to match multiple samples.

Several example data files from Byonic and MSFragger are provided which can be downloaded from the link in the screenshot below.


Upload Data File Hide

Purpose	Byonic	MSFragger	Customized
Glycan Composition ?	Glycan Composition	(empty)	(empty)
Protein Accessions	Protein Accessions	Protein	Protein
Position in Protein ?	Position in Protein	Protein Start	Start
Master Protein Descriptions	Master Protein Descriptions	Protein Description	Entry Name
Sequence	Sequence	Peptide	Peptide Sequence
Modifications ?	Modifications (all possible sites)	Assigned Modifications	Assigned Modifications
Retention Time	Top Apex RT [min]	Retention	nistmab1_1 Apex Retention Time
PREFIX-MATCHED COLUMNS			
FDR Prefix ?	FDR 2D (by Search Engine)	Probability	(empty)
Engine Score Prefix ?	Byonic Score (	Hyperscore	(empty)
PPM Prefix ?	DeltaM [ppm]	(empty)	(empty)
Abundance Columns ?	Abundances (Grouped)*	Intensity	*_1 Intensity

Saved

Upload an .xlsx, .tsv, or .csv file to begin a new analysis. (example file: byonic, msfragger_psm, msfragger_combined_ion)

Drag & drop your data file here, or click to browse
Accepted formats: .xlsx,.tsv,.csv
example_input_msfragger_psm (1).tsv

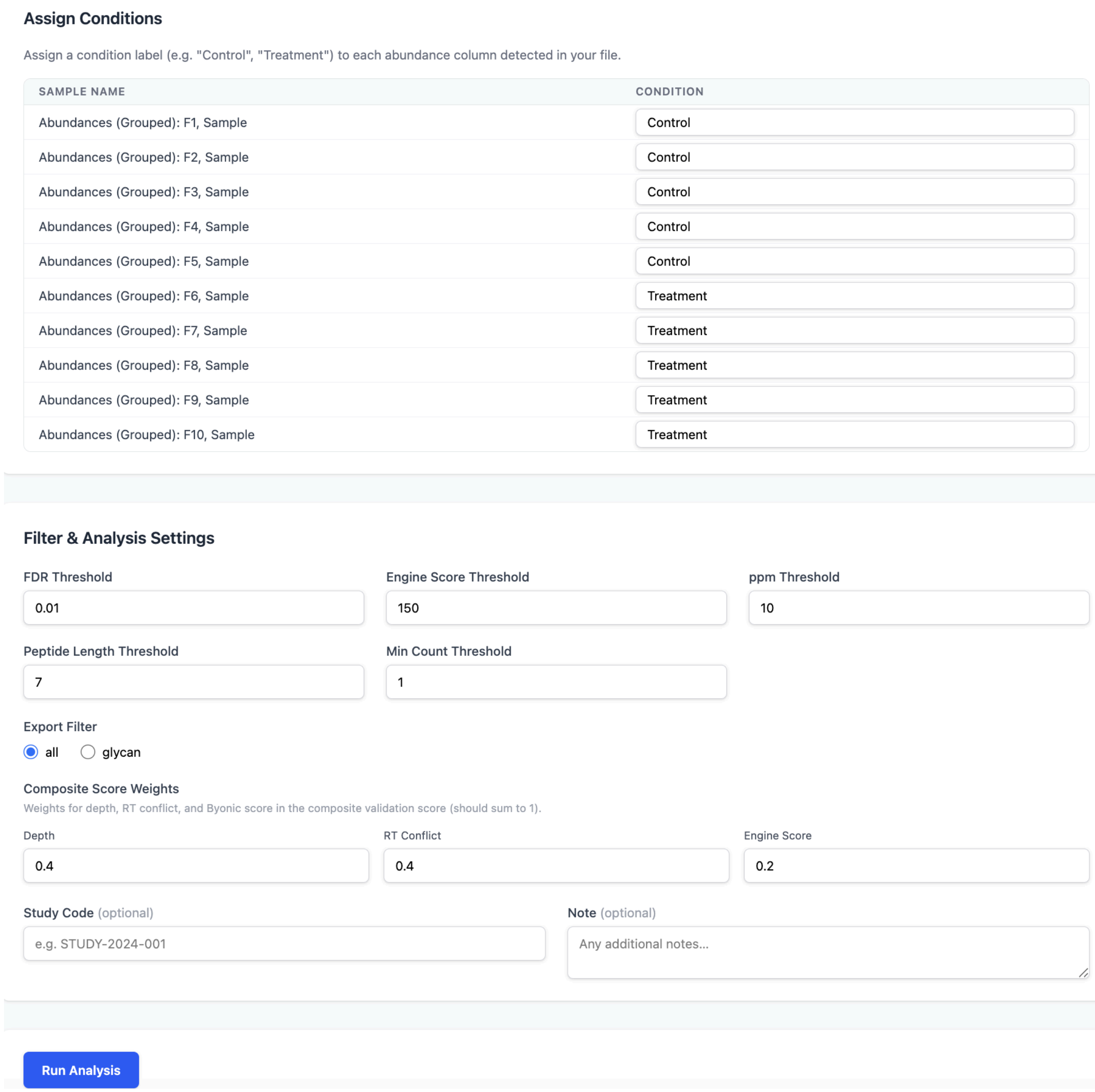
✓ Upload complete

✓ Detected format: **MSFragger**

1.3 Condition Assignment and Analysis Parameters

After uploading, GlycoViz displays the detected abundance columns. Assign each sample to a condition (e.g., "Control", "Treatment") and provide an analysis name. Adjust filters and click “Run Analysis” to begin processing.
If the dataset contains only a single condition, the volcano plot and heatmap will not load because they rely on differential analysis between conditions. .

“ Min Count Threshold” filters out glycans that don't appear in enough samples. Specifically:
- For each glycan row, it counts how many abundance columns have a positive value (i.e., how many samples detected that glycan).
- If that count is less than the threshold, the row is skipped/excluded from the analysis.



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2. Analysis Results

After analysis completes, click the analysis row in the Saved Analyses table to view results. The analysis page displays QC statistics (total proteins, peptides, glycopeptides, unique glycans, and sites) along with interactive visualizations.

2.1 Plot Controls

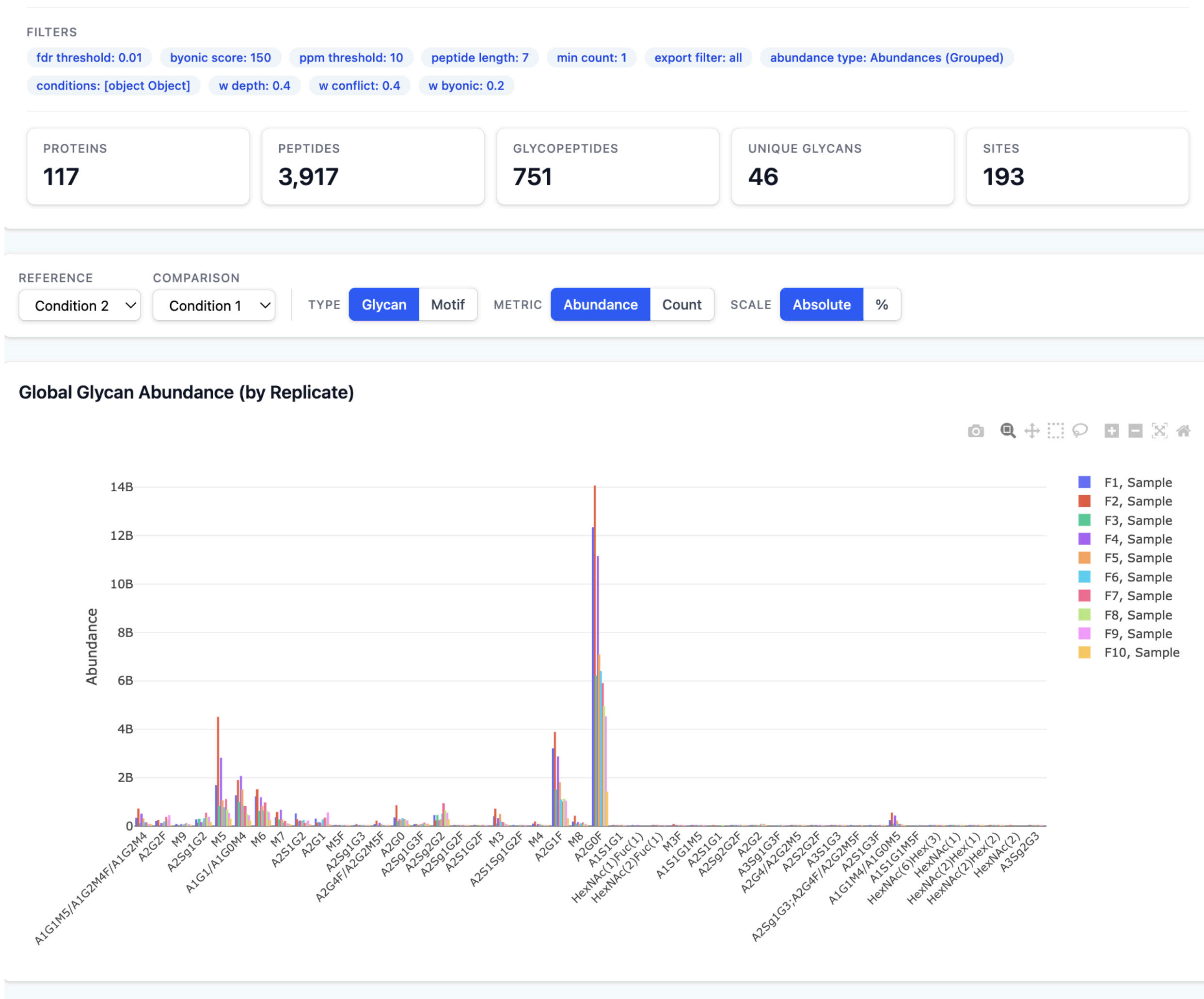
Three toggle groups at the top control all charts simultaneously:

- Type: Glycan or Motif — switch between individual glycan compositions and grouped motif categories (Fucosylation, High Mannose, Galactosylation, etc.)
- Metric: Abundance or Count — show summed intensities or spectral counts
- Scale: Absolute or % — display raw values or percentages of total

2.2 Global Abundance Bar Charts

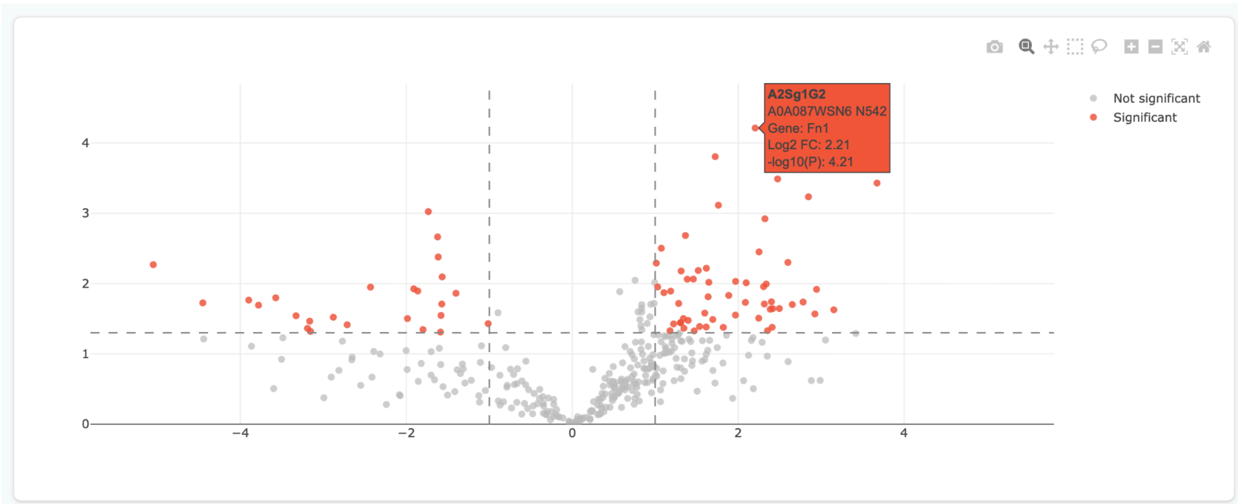
Two bar charts display glycan or motif abundance across all sites:

- By Replicate — each sample is a separate bar, showing raw per-sample values
- By Condition — samples are grouped by condition, showing mean ± standard error



2.3 Volcano Plot and Fold Change Analysis

This plot is only available if there are more than 1 condition defined when uploading the file. Select a reference and comparison condition to generate:



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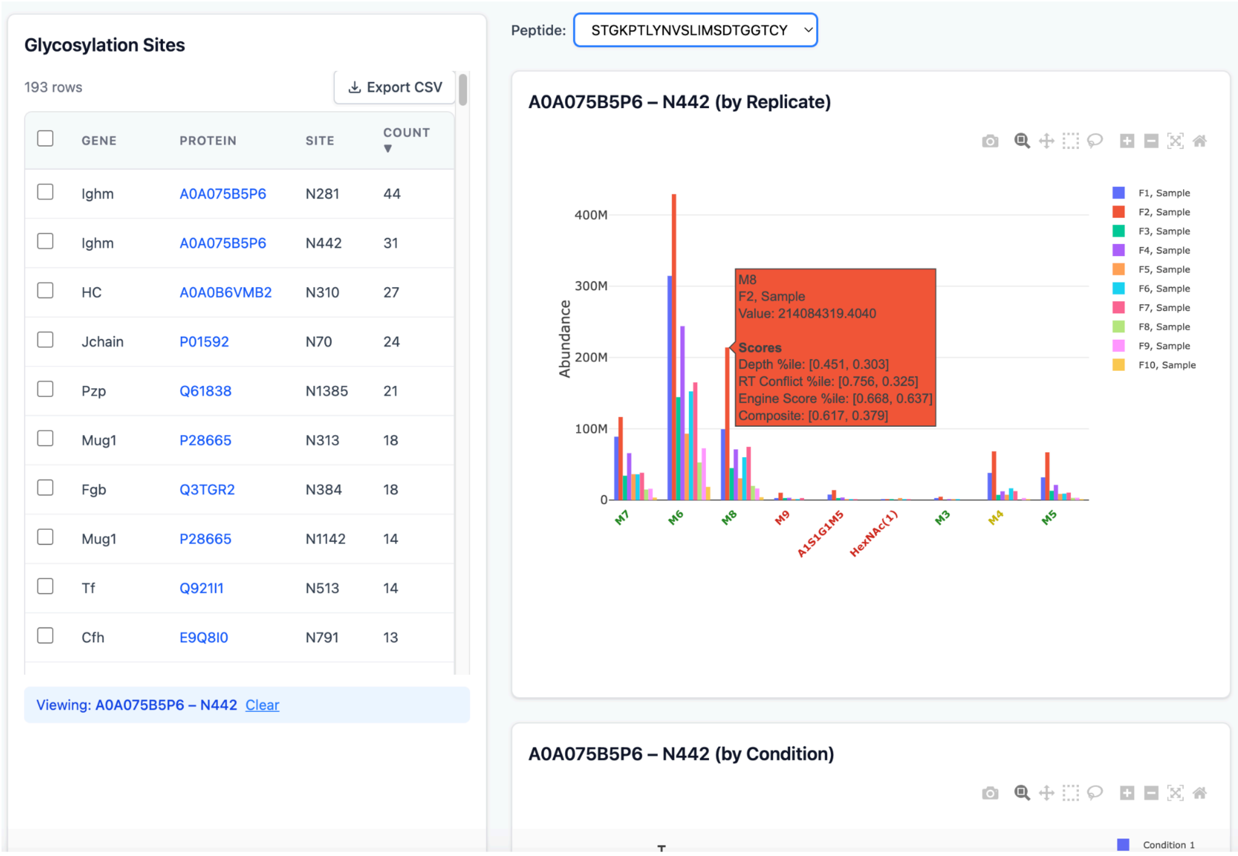
3. Site-Level Exploration

3.1 Glycosylation Sites Table

The left panel lists all identified glycosylation sites with gene name, protein accession, site position, and spectral count. Click any row to view per-site abundance charts in the right panel. Use checkboxes to select multiple sites, then click Display figures for all checked rows to view combined abundance data. The table is sortable and exportable to CSV.

3.2 Per-Site and Multi-Site Charts

Selecting a single site shows two bar charts (by replicate and by condition) specific to that glycosylation site, with composite validation score coloring. Selecting multiple sites via checkboxes shows combined abundance charts aggregated across all selected sites. Clicking a protein accession link navigates to the protein glycosylation map.
The detailed scoring matrix is available when hovering on a data point.



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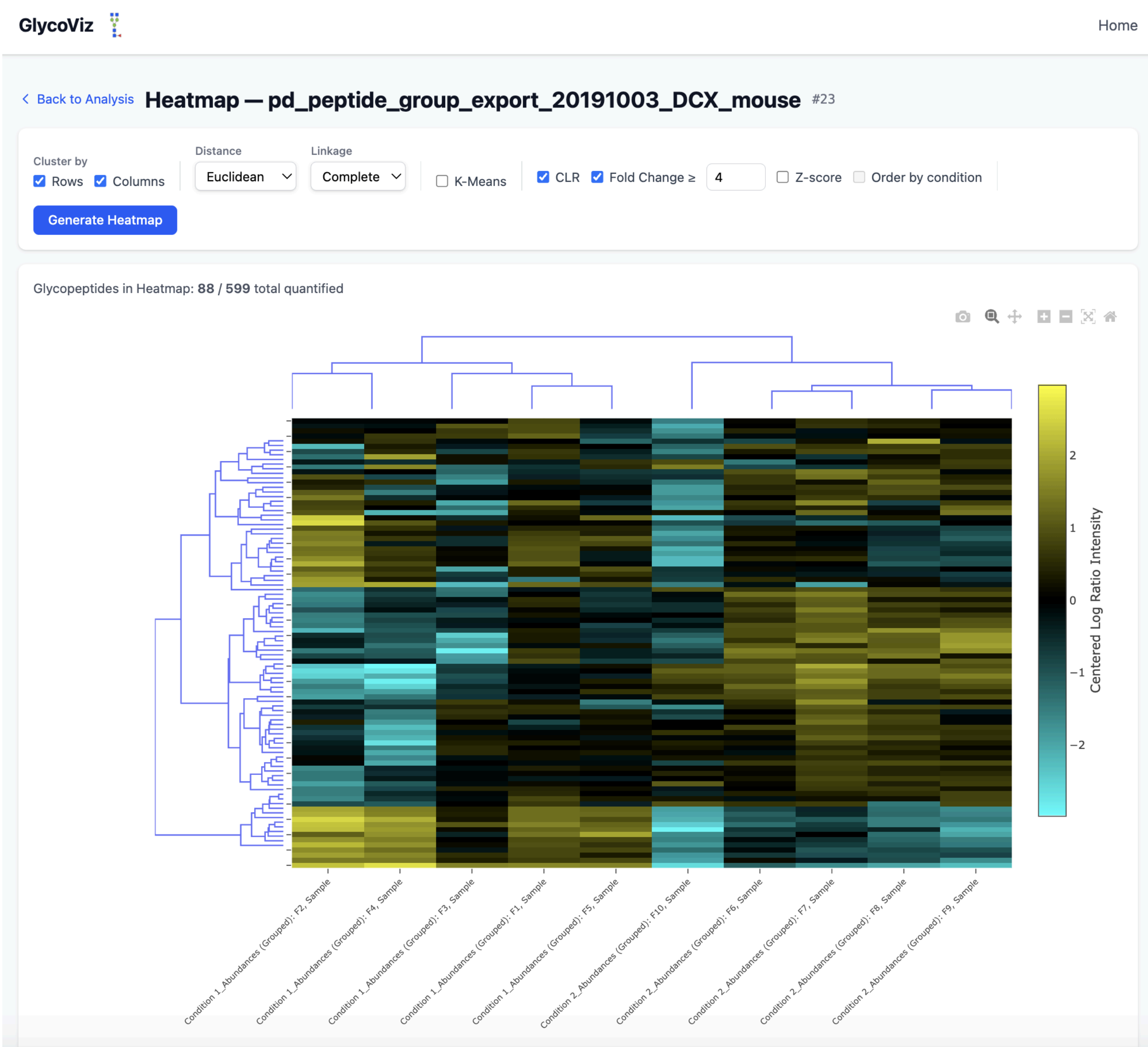
4. Heatmap and Clustering

Click View Heatmap from the analysis page to open the clustering visualization. Configure clustering parameters:

- Cluster by: rows, columns, or both
- Distance metric: Euclidean, Manhattan, Maximum, or Minkowski
- Linkage method: Complete, Single, or Average
- Transformations: Centered Log Ratio (CLR), Z-score normalization, and fold change filtering (minimum threshold, default ≥ 4)
- K-Means: optional k-means pre-clustering (k = 2–20)

Click Generate Heatmap to compute. The result displays an interactive heatmap with hierarchical dendrograms on rows and/or columns, using a cyan–black–yellow color scale centered at zero for log-ratio transforms.



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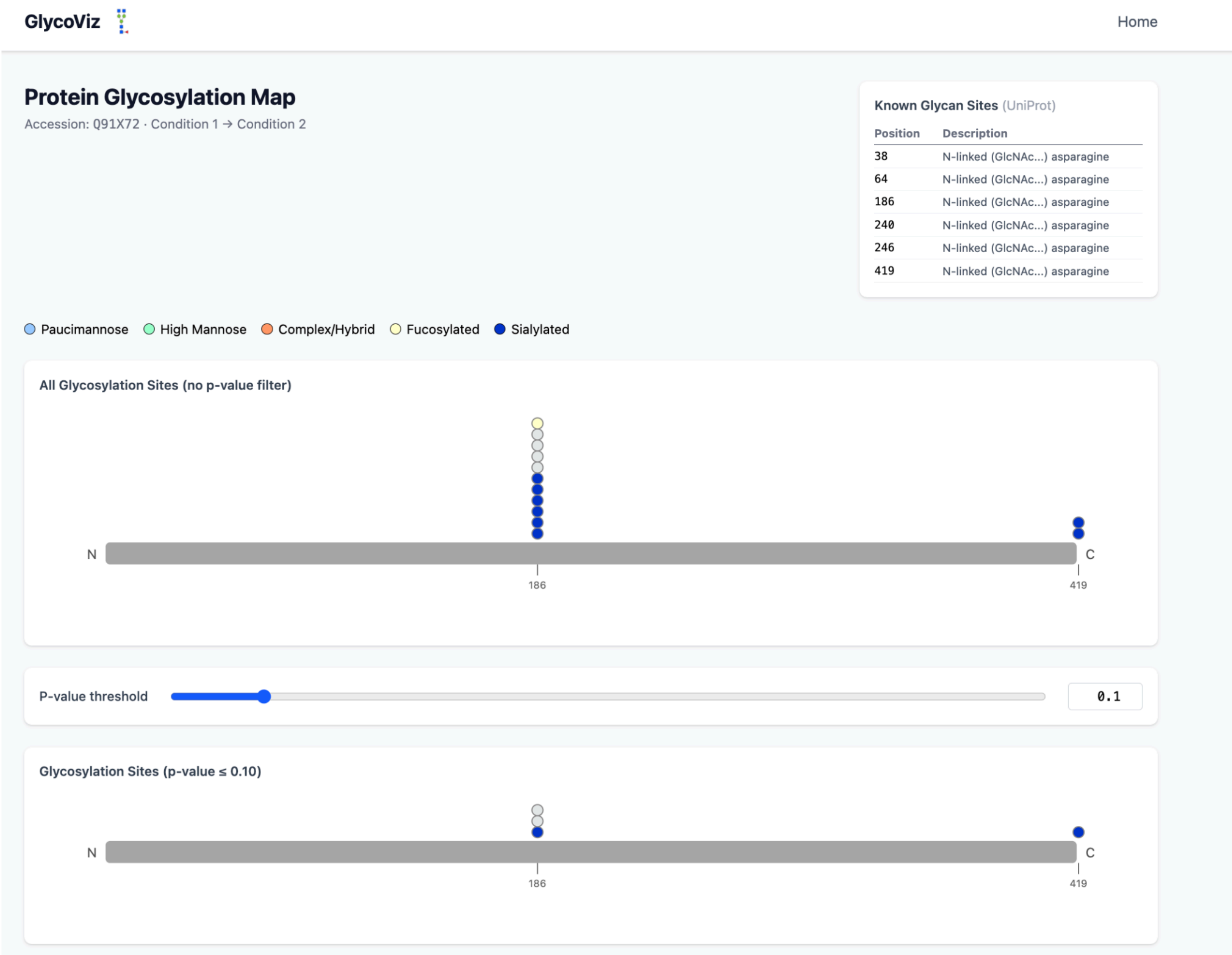
5. Protein Glycosylation Map

Click a protein accession name from the sites table to view the interactive glycosylation map. This page displays:

- An SVG diagram of the protein backbone with colored circles at each glycosylation site, color-coded by motif (High Mannose, Complex/Hybrid, Fucosylated, Sialylated, Paucimannose)
- Known glycosylation sites from UniProt (displayed in a sidebar table)
- A p-value filter slider to show only statistically significant sites



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6. Managing Analyses

The Saved Analyses table on the home page lists all completed analyses with name, study code, parameters, and creation date. Click any row to view results. Click the Parameter link to inspect all filter settings used. To delete an analysis and all associated files, click the Delete button at the end of row and type "DELETE" to confirm.

The application is provided "as is" without warranties of any kind.